Lab Task (30-03-2026)

CLO4

Mandatory Implementation Requirements

1. Multilevel Inheritance Structure

You must strictly follow:

Base Class → Intermediate Class → Derived Class

Example:

Media → Audio → Podcast

2. Common Behavior (Must be implemented in ALL tasks)

Each class must include the following **common behaviors**:

- **Core Methods (in Base Class)**
  - display() → prints all attributes of the object
  - toString() → returns formatted object details
  - At least **2 functional methods** depending on domain, e.g.:
    - start(), stop()
    - play(), pause()
    - open(), close()
    - attack(), defend()
- **Behavior Rules Across Classes**
  - Child classes must:

    - **override display()**
    - **override toString()**
- At least one method must:
    - call parent method using super.methodName()

3. Constructor Requirements

- All classes must have **parameterized constructors**
- Must use:

super(...)

- Use this keyword where needed

4. Access Modifier Rule

- All attributes must be declared using:

protected

5. Demo/Test Class (Compulsory)

You must create a separate class:

Demo.java

It must:

- Create **at least one object of each class**
- Call:
    - display()
    - toString()
    - other methods
- Clearly demonstrate:

- inheritance
- method overriding
- constructor chaining

GitHub Submission Requirement (VERY IMPORTANT)

This assignment must be submitted via **GitHub**.

Step 1: Create Repository

- Go to GitHub
- Create a new repository named:

OOP-Inheritance-[YourRegistrationNumber]

Example:

OOP-Inheritance-CIIT-FA25-BCS-003

Step 2: Upload Code

Your repository must contain:

```
/src
  BaseClass.java
  IntermediateClass.java
  DerivedClass.java
  Demo.java
```

◆ **Step 4: GitHub Link Submission**

After uploading your code:

- Copy your repository link

- Submit it on the **Google Form**

Google Form Submission : https://forms.gle/UG3fFGfmLNWN2cEh9

Each student must submit:

Required Fields:

1. Student Name
2. Registration Number
3. Task Number
4. GitHub Repository Link

Reg. No.	Student Name	Multilevel Inheritance Chain	Base Class Details	Intermediate Class Details	Derived Class Details	Demo Requirement
CIIT/FA25-BCS-003/LHR	AHMAD IQBAL	Media → Audio → Podcast	Media: fileSize, duration; methods: play(), pause(), stop(), display(), toString()	Audio: bitrate, artist; override display(), toString()	Podcast: hostName, episodeNumber; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-009/LHR	ALEEZA MUJEEB	Vehicle → Car → ElectricCar	Vehicle: brand, speed; methods: start(), stop(), display(), toString()	Car: numberOfDoors, fuelType; override display(), toString()	ElectricCar: batteryCapacity, chargingTime; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-015/LHR	AMNA MUSTAFA	Person → Student → GraduateStudent	Person: name, age; methods: walk(), eat(), display(), toString()	Student: registrationNumber, semester; override display(), toString()	GraduateStudent: researchTopic, supervisorName; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-021/LHR	BAREERAH NOOR	Employee → Manager → ProjectManager	Employee: employeeId, salary; methods: work(), display(), toString()	Manager: department, teamSize; override display(), toString()	ProjectManager: projectName, deadline; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-024/LHR	EISHAL ASIM	Product → ElectronicProduct → SmartPhone	Product: productId, price; methods: sell(), display(), toString()	ElectronicProduct: brand, warrantyYears; override display(), toString()	SmartPhone: ram, cameraMP; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-027/LHR	FAJR FATIMA	Book → EBook → AudioBook	Book: title, author; methods: open(), close(), display(), toString()	EBook: fileFormat, fileSize; override display(), toString()	AudioBook: narrator, duration; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-030/LHR	FIZA RASOOL	Account → SavingsAccount → PremiumSavingsAccount	Account: accountNumber, balance; methods: deposit(), withdraw(), display(), toString()	SavingsAccount: interestRate, minimumBalance; override display(), toString()	PremiumSavingsAccount: rewardPoints, freeTransactions; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-033/LHR	HABIBA AMJAD	Appliance → WashingMachine → SmartWashingMachine	Appliance: brand, powerRating; methods: turnOn(), turnOff(), display(), toString()	WashingMachine: capacity, washModes; override display(), toString()	SmartWashingMachine: wifiEnabled, appControl; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-039/LHR	ISBAH WAHEED	Shape → Rectangle → Box	Shape: color, borderWidth; methods: draw(), display(), toString()	Rectangle: length, width; override display(), toString()	Box: height, volume; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-045/LHR	MOAZZAM MUJAHID	HospitalMember → Doctor → Surgeon	HospitalMember: name, id; methods: checkIn(), display(), toString()	Doctor: specialization, experienceYears; override display(), toString()	Surgeon: surgeryType, operationRoomNumber; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-048/LHR	MUHAMMAD ABDULLAH	Ticket → BusTicket → ACBusTicket	Ticket: ticketNumber, fare; methods: book(), cancel(), display(), toString()	BusTicket: route, seatNumber; override display(), toString()	ACBusTicket: acLevel, snackIncluded; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-051/LHR	MUHAMMAD ABUZAR	GameCharacter → Warrior → Knight	GameCharacter: name, health; methods: attack(), defend(), display(), toString()	Warrior: weaponType, armorLevel; override display(), toString()	Knight: horseName, kingdom; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-054/LHR	MUHAMMAD BILAL KHAN	Building → House → SmartHouse	Building: address, area; methods: openGate(), closeGate(), display(), toString()	House: numberOfRooms, ownerName; override display(), toString()	SmartHouse: securitySystem, voiceControl; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-057/LHR	MUHAMMAD IBRAHIM	Course → LabCourse → ProgrammingLab	Course: courseCode, courseTitle; methods: startCourse(), endCourse(), display(), toString()	LabCourse: labNumber, labInstructor; override display(), toString()	ProgrammingLab: language, numberOfPCs; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-060/LHR	MUHAMMAD SAAD	Animal → Bird → Eagle	Animal: name, age; methods: eat(), sleep(), display(), toString()	Bird: wingSpan, canFly; override display(), toString()	Eagle: visionRange, clawStrength; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-066/LHR	NABIHA KHAWAR	Device → Computer → Laptop	Device: deviceId, brand; methods: powerOn(), powerOff(), display(), toString()	Computer: processor, ram; override display(), toString()	Laptop: batteryLife, weight; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-069/LHR	RAMEEN ZAHID	RestaurantItem → Meal → Dessert	RestaurantItem: itemName, price; methods: prepare(), serve(), display(), toString()	Meal: calories, spiceLevel; override display(), toString()	Dessert: sweetnessLevel, isColdServed; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-072/LHR	MUHAMMAD REHAN SAGHEER	Media → Audio → Podcast	Media: fileSize, duration; methods: play(), pause(), stop(), display(), toString()	Audio: bitrate, artist; override display(), toString()	Podcast: hostName, episodeNumber; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-075/LHR	SAHAR BATUL REHAN	Vehicle → Car → ElectricCar	Vehicle: brand, speed; methods: start(), stop(), display(), toString()	Car: numberOfDoors, fuelType; override display(), toString()	ElectricCar: batteryCapacity, chargingTime; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-084/LHR	AHMAD HASSAN	Person → Student → GraduateStudent	Person: name, age; methods: walk(), eat(), display(), toString()	Student: registrationNumber, semester; override display(), toString()	GraduateStudent: researchTopic, supervisorName; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-087/LHR	HAMZA KHALID	Employee → Manager → ProjectManager	Employee: employeeId, salary; methods: work(), display(), toString()	Manager: department, teamSize; override display(), toString()	ProjectManager: projectName, deadline; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-090/LHR	MAHEEN ZAHRA	Product → ElectronicProduct → SmartPhone	Product: productId, price; methods: sell(), display(), toString()	ElectronicProduct: brand, warrantyYears; override display(), toString()	SmartPhone: ram, cameraMP; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-099/LHR	SYEDA NOOR-E-ZAHRA BUKHARI	Book → EBook → AudioBook	Book: title, author; methods: open(), close(), display(), toString()	EBook: fileFormat, fileSize; override display(), toString()	AudioBook: narrator, duration; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-102/LHR	MUHAMMAD ABDULLAH	Account → SavingsAccount → PremiumSavingsAccount	Account: accountNumber, balance; methods: deposit(), withdraw(), display(), toString()	SavingsAccount: interestRate, minimumBalance; override display(), toString()	PremiumSavingsAccount: rewardPoints, freeTransactions; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-105/LHR	RABBIA SAIF	Appliance → WashingMachine → SmartWashingMachine	Appliance: brand, powerRating; methods: turnOn(), turnOff(), display(), toString()	WashingMachine: capacity, washModes; override display(), toString()	SmartWashingMachine: wifiEnabled, appControl; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-108/LHR	AHMED JAVED	Shape → Rectangle → Box	Shape: color, borderWidth; methods: draw(), display(), toString()	Rectangle: length, width; override display(), toString()	Box: height, volume; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-111/LHR	ISHMAL AMAN	HospitalMember → Doctor → Surgeon	HospitalMember: name, id; methods: checkIn(), display(), toString()	Doctor: specialization, experienceYears; override display(), toString()	Surgeon: surgeryType, operationRoomNumber; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-114/LHR	MUHAMMAD EISSA DAR	Ticket → BusTicket → ACBusTicket	Ticket: ticketNumber, fare; methods: book(), cancel(), display(), toString()	BusTicket: route, seatNumber; override display(), toString()	ACBusTicket: acLevel, snackIncluded; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-117/LHR	KHADIJA KASHIF	GameCharacter → Warrior → Knight	GameCharacter: name, health; methods: attack(), defend(), display(), toString()	Warrior: weaponType, armorLevel; override display(), toString()	Knight: horseName, kingdom; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-120/LHR	EMAAN AHMAD	Building → House → SmartHouse	Building: address, area; methods: openGate(), closeGate(), display(), toString()	House: numberOfRooms, ownerName; override display(), toString()	SmartHouse: securitySystem, voiceControl; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-126/LHR	AHMAD RASHID	Course → LabCourse → ProgrammingLab	Course: courseCode, courseTitle; methods: startCourse(), endCourse(), display(), toString()	LabCourse: labNumber, labInstructor; override display(), toString()	ProgrammingLab: language, numberOfPCs; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-132/LHR	YASHFA ZAHID	Animal → Bird → Eagle	Animal: name, age; methods: eat(), sleep(), display(), toString()	Bird: wingSpan, canFly; override display(), toString()	Eagle: visionRange, clawStrength; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-138/LHR	AMNA USMAN	Device → Computer → Laptop	Device: deviceId, brand; methods: powerOn(), powerOff(), display(), toString()	Computer: processor, ram; override display(), toString()	Laptop: batteryLife, weight; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-141/LHR	ZAIN RAZA	RestaurantItem → Meal → Dessert	RestaurantItem: itemName, price; methods: prepare(), serve(), display(), toString()	Meal: calories, spiceLevel; override display(), toString()	Dessert: sweetnessLevel, isColdServed; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-144/LHR	HAFSA AFZAL	Media → Audio → Podcast	Media: fileSize, duration; methods: play(), pause(), stop(), display(), toString()	Audio: bitrate, artist; override display(), toString()	Podcast: hostName, episodeNumber; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-147/LHR	MOMIN	Vehicle → Car → ElectricCar	Vehicle: brand, speed; methods: start(), stop(), display(), toString()	Car: numberOfDoors, fuelType; override display(), toString()	ElectricCar: batteryCapacity, chargingTime; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-153/LHR	SABAHAT AKBAR	Person → Student → GraduateStudent	Person: name, age; methods: walk(), eat(), display(), toString()	Student: registrationNumber, semester; override display(), toString()	GraduateStudent: researchTopic, supervisorName; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-156/LHR	MUHAMMAD RABEE UMAR	Employee → Manager → ProjectManager	Employee: employeeId, salary; methods: work(), display(), toString()	Manager: department, teamSize; override display(), toString()	ProjectManager: projectName, deadline; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-159/LHR	BISMA USMAN KHAN	Product → ElectronicProduct → SmartPhone	Product: productId, price; methods: sell(), display(), toString()	ElectronicProduct: brand, warrantyYears; override display(), toString()	SmartPhone: ram, cameraMP; override display(), toString()	Create objects of all classes and call all methods

CIIT/FA25-BCS-162/LHR	MUHAMMAD SAQIB	Book → EBook → AudioBook	Book: title, author; methods: open(), close(), display(), toString()	EBook: fileFormat, fileSize; override display(), toString()	AudioBook: narrator, duration; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-167/LHR	MUHAMMAD MUFFASER NAEEM	Account → SavingsAccount → PremiumSavingsAccount	Account: accountNumber, balance; methods: deposit(), withdraw(), display(), toString()	SavingsAccount: interestRate, minimumBalance; override display(), toString()	PremiumSavingsAccount: rewardPoints, freeTransactions; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-168/LHR	ABDUL HAMEED QAISAR	Appliance → WashingMachine → SmartWashingMachine	Appliance: brand, powerRating; methods: turnOn(), turnOff(), display(), toString()	WashingMachine: capacity, washModes; override display(), toString()	SmartWashingMachine: wifiEnabled, appControl; override display(), toString()	Create objects of all classes and call all methods
CIIT/FA25-BCS-171/LHR	FATIMA LIAQUAT	Shape → Rectangle → Box	Shape: color, borderWidth; methods: draw(), display(), toString()	Rectangle: length, width; override display(), toString()	Box: height, volume; override display(), toString()	Create objects of all classes and call all methods